

Prof. Dr. Byron Jacobs

Chief Science Officer

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Professional Profile

Applied Mathematician and ML Expert with over 14 years of experience specializing in the deployment of emerging technologies. Skilled in building varied ML based solutions and solving high-stakes industrial problems through rigorous scientific methodology. Demonstrated success in leading R&D initiatives, managing multi-stakeholder projects, and fostering knowledge transfer within technical organizations.

Core Expertise

- **Applied Mathematics & AI:** Data-driven modelling, LLM-based solutions, Machine Learning for Medtech, and predictive analytics.
- **Software & Development:** Algorithm design and implementation in Python, Mathematica, and MATLAB, version control and CI/CD.
- **Cloud & Digital Strategy:** Architecture and deployment on AWS and GCP, data-driven decision-making frameworks.
- **Technical Leadership:** Management of cross-functional teams, R&D project oversight, curriculum innovation, and stakeholder communication.

Professional Experience

Chief Science Officer & Co-Founder

VitruvianMD

Jan 2020 – Present

Johannesburg, ZA

- **Innovation:** Designed and patented a novel predictive diagnostic platform for sperm DNA fragmentation using advanced AI/ML (PyTorch, TensorFlow, Hugging Face).
- **Global Scale:** Directed the deployment of AI-driven clinical solutions across five countries, managing international data privacy standards and ensuring technical scalability for concurrent users/clinics.
- **Strategic Funding:** Principal Investigator for the SAMRC / Bill & Melinda Gates Foundation LLM Grand Challenge.
- **Leadership:** Scaled and mentored a multidisciplinary engineering team, transitioning R&D concepts into commercially viable products.
- **Thought Leadership:** Authored high-impact white papers and peer-reviewed research; delivered strategic reports to investors and external stakeholders.

Associate Professor | Dept. of Mathematics & Applied Mathematics

University of Johannesburg

July 2020 – March 2024

Johannesburg, ZA

- **National Recognition:** Distinguished as an NRF Rated Scientist (Y2) for significant contributions to the field.
- **Talent Pipeline:** Supervised 3 PhD candidates and successfully established a strategic industry-engagement platform for Science & Engineering students.
- **Research Impact:** Maintained a consistent output of high-impact journal publications focusing on applied mathematical modeling. Peer-reviewer for over 20 top-tier journals in AI, Signal Processing, and Applied Mathematics, maintaining a pulse on the global frontier of emerging technology.
- **Global Integration:** Cultivated high-value international collaborations with top-tier institutions (UNSW, Linköping), serving as a catalyst for globally recognised research.
- **Enterprise Liaison:** Served as lead industry liaison for major financial institutions, facilitating technical innovation days and pre-interview testing for high-level engineering talent.

Associate Professor (Promoted from Lecturer)

University of the Witwatersrand

June 2012 – June 2020

Johannesburg, ZA

- **Rapid Advancement:** Accelerated from Associate Lecturer to Associate Professor within 6 years based on research excellence and leadership through post graduate supervision.
- **Funding & Grants:** Secured competitive NRF CSUR and Thuthuka grants to fund large-scale computational research (over R7.5m awarded across multiple grants).
- **Mentorship:** Directed the research of 14 MSc and PhD students, fostering a culture of rigorous analytical inquiry.
- **Strategic Design:** Led curriculum development for the Computational and Applied Mathematics degree to align academic output with modern industry requirements.

Education

PhD in Computational and Applied Mathematics (2014) *University of the Witwatersrand*
BSc Hons (cum laude) in Computational and Applied Mathematics (2010) *University of the Witwatersrand*
Rand Merchant Bank Gold Medal and the Starfield Prize for Best Modelling Project. (2010)

Selected Publications and Patents

- **Jacobs, B. A., Lin, P.J., Shaik, I.** (2025) "Visual Analysis of Sperm DNA Fragmentation." *US Patent App. 19/289,222*.
- **Singh, K., Jacobs, B. A.** (2025) "A Network Based Model for Predicting Spatial Progression of Metastasis." *Bulletin of Mathematical Biology*.
- **Mtshali, S. Jacobs, B. A.** (2025). "Machine learning-based prediction of pharmacokinetic parameters for individualized drug dosage optimization." *International Journal of Information Technology*.
- **Jacobs, B. A.** (2024). "Image trinarization using a partial differential equation: A novel approach to automatic sperm image analysis." *Applied Mathematical Modelling*.
- **Mtshali, S. Jacobs, B. A.** (2023). "On the validation of a fractional order model for pharmacokinetics using clinical data." *Fractal and Fractional*.
- **Angstmann, C. N., Donnelly, I.C., Henry, B.I., Jacobs, B. A., Langlands, T.A.M.** (2016). "From stochastic processes to numerical methods: A new scheme for solving reaction subdiffusion fractional partial differential equations." *Journal of Computational Physics*.
- **Jacobs, B. A., Momoniat, E.** (2013). "A novel approach to text binarization via a diffusion-based model." *Applied Mathematics and Computation*.